

SIRT2 and "defense response to virus" (GO:0051607): Restriction vs. Dependency

Gene: SIRT2 (human) · **UniProt:** Q8IXJ6 · **Taxon:** NCBITaxon:9606 **Focus:** function_assignment · **Seed hypothesis:** *SIRT2 has defense response to virus (GO:0051607)* **Deliverable:** direction verdict + decisive experiments + contrary evidence

Executive Judgment

Verdict: REFUTED as an unqualified assignment / OVER-ANNOTATION. Direction is VIRUS-SPECIFIC, and the preponderance of decisive perturbation evidence points to SIRT2 acting as a host DEPENDENCY (pro-viral) factor — the opposite sign to GO:0051607.

A single, unqualified "defense response to virus" annotation on human SIRT2 is **not warranted**. The sign of SIRT2's effect on virus replication is not uniform, and where the readout is *infectious progeny titre* (the decisive metric), SIRT2 behaves as a factor that viruses **require**, not one that restricts them:

- **Both decisive infectious-progeny studies point pro-viral.** In dengue, a SIRT2-directed inhibitor reduces release of infectious progeny and the effect is abolished by SIRT2 knockdown (on-target;

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In HIV-1, a SIRT2 inhibitor reduces viral growth by >1 log₁₀ in macrophages and humanized mice

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- **HBV (6 studies, multiple perturbation types)** consistently shows SIRT2 *promotes* replication: overexpression increases HBV nucleic acids

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and three independent chemical inhibitors/modulators (AGK2 —

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FLS-359 —

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suppress HBV.

- **HSV-1:** SIRT2 *suppresses* the antiviral cGAS-STING/type-I-IFN axis, so SIRT2 loss or inhibition *increases* interferon and clears virus

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— again pro-viral.

Contrary evidence exists but is weaker for GO:0051607. The antiviral (restriction-like) direction rests on (i) influenza, where SIRT2 *activation* lowers replication via redox restoration

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— an activation gain-of-function with a surrogate readout, not loss-of-function; (ii) SIRT2 support of IFN-stimulated-gene transcription via CDK9/STAT1

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— a signaling surrogate, no viral titre; and (iii) a minor alternatively spliced isoform 5 that inhibits HBV

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— isoform-specific and opposite to the dominant isoform 1.

Key caveat / important curation fact: A QuickGO query of all 206 GO annotations on Q8IXJ6 found **no existing viral-defense annotation** (GO:0051607 and all related/descendant viral terms return 0 hits). The seed is therefore a **proposed addition**, and the evidence does not support adding it in unqualified form.

Evidence Matrix

Citation	Evidence type	Stance vs seed	Claim tested	Key finding	Context
P 35663860	Direct assay / overexpression	Refutes	Does SIRT2 promote HBV?	Ectopic SIRT2 ↑ HBV RNAs, 3.5-kb RNA, core DNA (via p53)	HBV-infected HepG2-NTCP
P 29366781	Overexpression / pathway	Refutes	HBx→SIRT2→replication	SIRT2 facilitates HBV transcription/replication; HBx upregulates SIRT2	Hepatoma cells
P 30111572	Direct assay (isoform 1)	Refutes	SIRT2 isoform 1 vs HBV	Isoform 1 enhances HBV via AKT/GSK-3β/β-catenin	Hepatoma cells
P 30275764	Small-molecule inhibition	Refutes	Does SIRT2 inhibition ↓ HBV?	AGK2 inhibits HBV replication	HBV cell model
P 40270769	Small-molecule inhibition	Refutes	AGK2 epigenetic anti-HBV	AGK2 ↓ SIRT2, ↓ HBV RNA/DNA, ↓ cccDNA (repressive histone marks)	HBV-transfected/infected cells
P 38641024	Allosteric inhibition	Refutes	SIRT2 needed for cccDNA?	FLS-359 (SIRT2 modulator) blocks cccDNA establishment & transcription	HBV infection model

Citation	Evidence type	Stance vs seed	Claim tested	Key finding	Context
P 34147476	Inhibitor + siRNA (rescue)	Refutes	SIRT2 required for DENV?	Tenovin-1 ↓ infectious progeny release ; effect abolished by SIRT2 knockdown (on-target)	DENV1–4, BHK-21/Vero
P 41883165	Small-molecule inhibition	Refutes	SIRT2 required for HIV-1?	Sirtinol ↓ HIV-1 growth >1 log ₁₀	Human macrophages; humanized mice
P 36719240	Association / biomarker	Refutes	SIRT2 vs HIV load	Plasma SIRT2 ↑ with viral loads & provirus levels	People living with HIV
P 37870259	KO + inhibitor + interaction	Refutes	SIRT2 vs antiviral IFN	SIRT2 deacetylates G3BP1, suppresses cGAS-STING; deficiency ↑ IFN; AGK2 protects mice from HSV-1	HSV-1; mouse; cell
P 40220296	Mutant phenotype	Qualifies	SIRT2 in COVID severity	SIRT2 suppresses aging cGAS; SIRT2-KO aged mice get <i>severe</i> COVID	Aged mice, SARS-CoV-2
P 35071051	Sirtuin activation	Supports	Does SIRT2 restrict IAV?	SIRT2 activator	Influenza A, cell culture

Citation	Evidence type	Stance vs seed	Claim tested	Key finding	Context
				restores G6PD/ NRF2/GSH redox, ↓ IAV replication; virus downregulates SIRT2	
P 30487288	Signaling assay	Supports	SIRT2 in IFN response	SIRT2 required for ISG transcription via CDK9 deacetylation / STAT1-Ser727	MEFs / cells
P 32493816	Isoform assay	Supports (isoform)	Isoform 5 vs HBV	Alt-spliced SIRT2 isoform 5 <i>inhibits</i> HBV from cccDNA (repressive epigenetics)	HBV cell model

Tally: 10 refute (pro-viral/dependency), 3 support (all activation-based, signaling-surrogate, or minor-isoform), 1 qualifies (anti-inflammatory protection, not restriction). Provenance: evidence-matrix code + output executed in Iteration 2; CSV saved to the code sandbox at `/tmp/sirt2_virus_evidence_matrix.csv`.

GO Curation Implications (leads requiring curator verification)

- **GO:0051607 (defense response to virus, BP): Do NOT add as an unqualified annotation.** It is not currently in the GO record for Q8IXJ6 (QuickGO: 0 hits), and the primary literature does not support a uniform host-restriction role. If any viral statement is captured at all, it should be **virus-qualified** and directionally correct.

- The **dominant, mechanistically coherent** signal (HBV, dengue, HIV-1, HSV-1) is that SIRT2 is **pro-viral / permissive**. If a curator wishes to record this, the accurate directional term is closer to **positive regulation of viral genome replication (GO:0045070)** or **negative regulation of type I interferon-mediated / innate immune response** (e.g., via GO:0032480 negative regulation of type I IFN production), each **paper- and virus-specific with the appropriate NOT/qualifier discipline** — **not** GO:0051607.
- **Preferred core annotations remain the well-supported ones:** NAD⁺-dependent protein/ α -tubulin deacetylase activity (MF), protein deacetylation (GO:0006476, BP, IDA), and cytoplasm/microtubule/centrosome localization (CC). The antiviral phenotypes are **downstream, context-dependent consequences** of this deacetylase activity, not a distinct "defense response to virus" function.
- **Isoform caution:** Any HBV-restriction statement

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 is **isoform-5-specific** and should not be propagated to the canonical Q8IXJ6 entry.

Mechanistic Scope

The **immediate molecular function** being modulated in every study is SIRT2's **NAD⁺-dependent lysine deacetylase activity** (substrates include α -tubulin, G3BP1, CDK9, G6PD, p53, β -catenin pathway components). Virus outcomes are **downstream** of this enzyme activity:

- Pro-viral routes: SIRT2 deacetylates **G3BP1** to disassemble the cGAS-G3BP1 condensate and dampen STING/IFN (HSV-1, COVID); supports **AKT/GSK-3 β /** β -catenin and **p53**-dependent enhancer activity for HBV; is co-opted by dengue and HIV-1.
- Antiviral routes: SIRT2 deacetylates **CDK9** to promote STAT1-Ser727/ISG transcription; deacetylates **G6PD** to sustain NADPH/GSH redox that limits influenza.

Thus "defense response to virus" is at best a **pleiotropic, context-specific downstream phenotype**, not a primary gene-product function.

Conflicts and Alternatives

- **Genuine biological heterogeneity**, not artifact: same enzyme, opposite outcomes depending on which acetyl-substrate/pathway dominates in a given virus/cell system.
- **Perturbation-type asymmetry**: antiviral effects rely on **SIRT2 activation** or **signaling surrogates**; pro-viral effects are shown by **inhibition + genetic knockdown with on-target rescue** (dengue) and **in vivo** inhibition (HIV-1) — methodologically stronger for settling the sign.
- **Isoform confounding**: isoform 1 (pro-HBV) vs isoform 5 (anti-HBV) from the same locus.
- **Paralog caution**: many "sirtuin" antiviral claims involve **SIRT1** or pan-sirtuin tools (e.g., tenovin-1 is SIRT1/2; resveratrol/SRT compounds activate SIRT1). Curators must avoid SIRT1→SIRT2 carry-over.
- **Phenotype vs restriction**: SARS-CoV-2 protection

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 is **reduced immunopathology/disease severity**, not reduced viral replication — a different GO concept from GO:0051607.

Knowledge Gaps

1. **Clean genetic loss-of-function on infectious titre.** Most decisive HBV/HIV data use inhibitors; only dengue pairs inhibitor with siRNA. *Checked*: no CRISPR-KO study reporting infectious titre increase upon SIRT2 loss. *Resolve*: isogenic SIRT2-KO vs WT, plaque/TCID50 across virus panel.
2. **Catalytic dependence.** *Checked*: inhibitor data imply catalytic requirement but few catalytic-dead (H187Y) rescues. *Resolve*: WT vs catalytic-dead re-expression in KO.
3. **Isoform-resolved effects.** *Resolve*: isoform-specific knockdown/expression with titre readout.
4. **Inhibitor off-target / SIRT1 contribution.** *Resolve*: SIRT2-selective genetic tools alongside chemistry.
5. **Whether any virus is truly restricted by endogenous SIRT2 (LoF ↑ titre).** *Checked*: not found for influenza (only activation GoF). *Resolve*: SIRT2 depletion + IAV plaque assay.

Discriminating Tests

- **Isogenic SIRT2-KO (CRISPR) vs WT**, infectious-progeny titre (plaque/TCID50), across DENV, HIV-1, HBV, HSV-1, IAV — the single most decisive experiment for the sign.
- **Catalytic-dead rescue (H187Y)** to tie any effect to deacetylase activity.
- **SIRT2-selective inhibitor vs pan-sirtuin tool** side-by-side to exclude SIRT1.
- **Isoform-specific reconstitution** (isoform 1 vs 5) in KO cells.
- **G3BP1/cGAS-STING reporter** $\pm$ SIRT2 to confirm the immunosuppressive (pro-viral) axis quantitatively.

Curation Leads (require curator verification)

- **Action:** Reject/withhold unqualified GO:0051607 on Q8IXJ6 (**over-annotation**). If a viral role is captured, make it **virus-specific and directionally correct** (predominantly pro-viral / negative regulation of innate antiviral response).

- **Candidate references + snippets to verify:**

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— *"the release of progeny viruses was inhibited by tenovin-1 treatment" and "The antiviral effect of tenovin-1 was abrogated with SIRT agonist or SIRT2 knockdown treatment, which indicated the effect of tenovin-1 was on-target"* (decisive, pro-viral).

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— *"Sirtinol, a Sirt2 inhibitor, significantly reduced HIV-1 growth in M0, M1, and M2-MΦs by >1log10 over 7 days"* (pro-viral, in vivo).

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— *"ectopic SIRT2 overexpression markedly increased total HBV RNAs, 3.5-kb RNA and HBV core DNA"* (pro-viral).

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— *SIRT2 suppresses cGAS-STING; "SIRT2 deficiency markedly elevated the expression levels of type I interferon"* (pro-viral/immunosuppressive).

- Contrary (to weigh, not to base GO:0051607 on):

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(IAV, activation),

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(ISG signaling),

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(isoform 5).

- **Candidate replacement/qualified terms:** positive regulation of viral genome replication (GO:0045070); negative regulation of type I interferon production (GO:0032480) / negative regulation of innate immune response — each paper- and virus-scoped. Retain core MF (protein/tubulin deacetylase) and CC (cytoplasm/microtubule).
- **Suggested curator questions:** Is the proposed GO:0051607 from text-mining/paralog inference? Does any cited paper show LoF *increasing* infectious titre? Are isoform and SIRT1 confounds controlled?

Limitations

- Assessment is literature- and database-driven (PubMed + QuickGO); no wet-lab replication.
- Several HBV/HIV readouts are surrogates (nucleic acid, growth) rather than plaque titre.
- Literature coverage, while broad (HBV, HIV-1, dengue, HSV-1, IAV, SARS-CoV-2), is not exhaustive; additional virus-specific reports may exist.

Direction verdict: Virus-specific, dependency-dominant. Decisive infectious-progeny evidence (dengue

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HIV-1

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and the HBV/HSV-1 bodies of work place SIRT2 as a **host dependency / pro-viral factor**, contradicting an unqualified "defense response to virus" assignment. Restriction-like observations are real but confined to specific contexts (influenza via activation; IFN/ISG signaling; a minor HBV isoform) and do not justify GO:0051607 for canonical human SIRT2.

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